Module Introduction

Dr. Hongping Cai

CM22009 – Machine Learning

This lecture adapts material originally prepared by Dr. Andy Barnes (with permission)

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Teaching Team

Dr. Hongping Cai (Unit Convenor)
Lecturer
- Computer Vision
- Machine Learning
- Feature Learning

Email: hc551@bath.ac.uk

Dr. Georgios Exarchakis
Lecturer
- Representation Learning
- Efficient Probabilistic Inference
- Physics Informed Neural Networks

Email: ge394@bath.ac.uk

Dr. Deblina Bhattacharjee
Lecturer
- Interdisciplinary AI researcher
- Computer Vision and Visual Arts on generative models
- Multitask learning

Email: db2466@bath.ac.uk

Main mode of contact: ***Moodle Discussion Forum***

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Logistics

- All year (S1 & S2)
- Lectures
  - Monday 13:15 Edge Theatre
  - Wednesday 11:15 Edge Theatre
- Lab (please turn up to your allocated lab)
  - Starts from Week 1

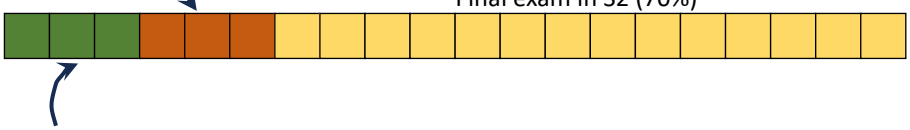
Group A:	3.15pm Thursday	Library Level 4 GTA PC area	Dina Theodosiadou Joe Hartley Viraj Patel Benjamin Mcdowell
Group B:	11.15am Friday	Library Level 4 GTA PC area	Dina Theodosiadou Joe Hartley Viraj Patel Marcel Venturotti
Group C:	9.15am Friday	EB 0.7 [Comp Rm]	Joe Hartley Balazs Nyiro

3

Assessment

Coursework in S2 – Experimental Report (15%)

Final exam in S2 (70%)



Coursework in S1 – Experimental Poster (15%)
- Date of Announcement: 22 October (Wed.), 2025
- Due date: **8pm, 14 November (Fri.), 2025**

You need both the theory and practice to pass this course.
- This will also be what you need to find a job in industry!

*Note: All questions about the coursework will be answered in the **Moodle Discussion Forum**, then every one can benefit from the answers.*

4

What is this course?

Machine Learning

5

What is this course?

Alan Turing
"… machines must learn in the same way as a human child."

Machine Learning

6

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Machine Learning

Arthur Samuels
First use of the term 'machine learning' when creating a program which learnt to play checkers.

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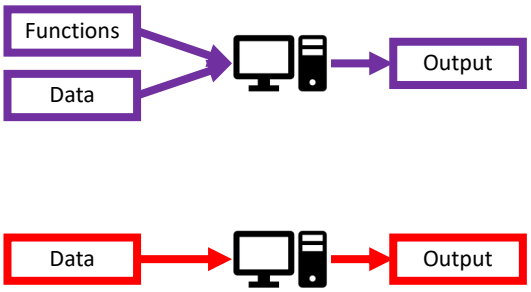

9

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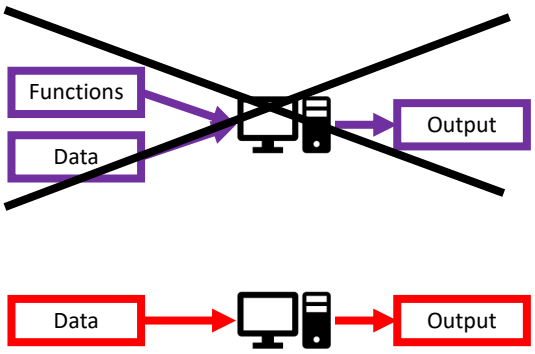
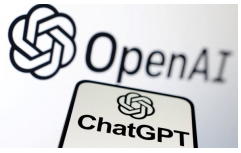
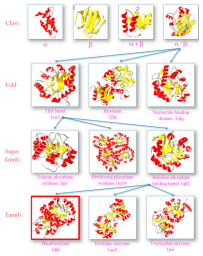


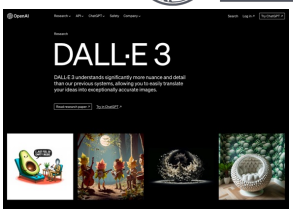
ML: Algorithms to learn rules from examples

10

What is this course?







Machine Learning

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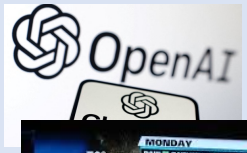
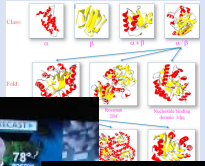
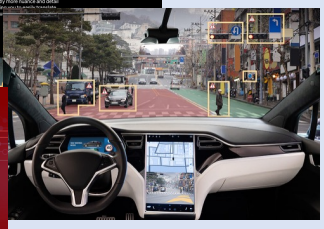
What is this course?

Artificial Intelligence

Machine Learning







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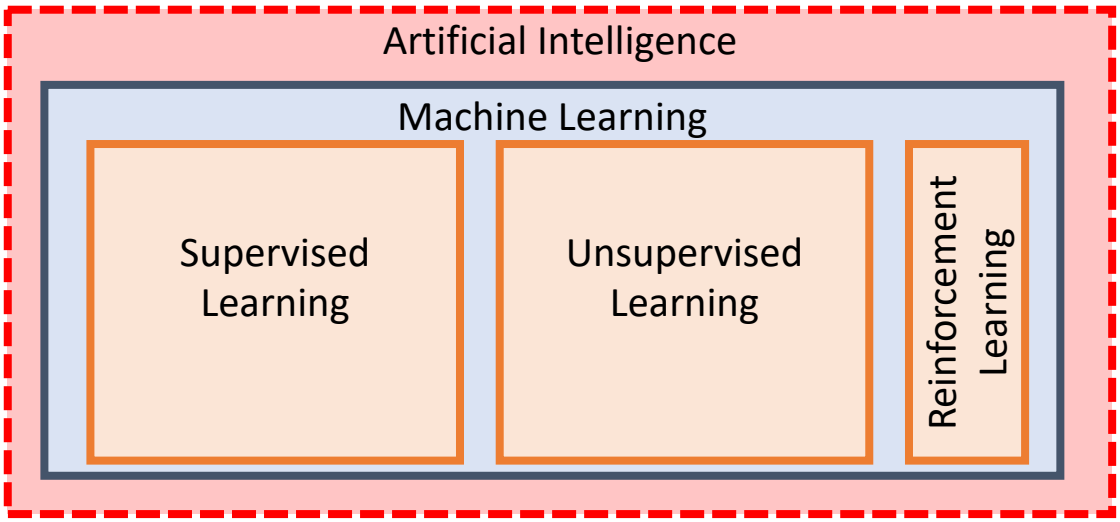
What is this course?



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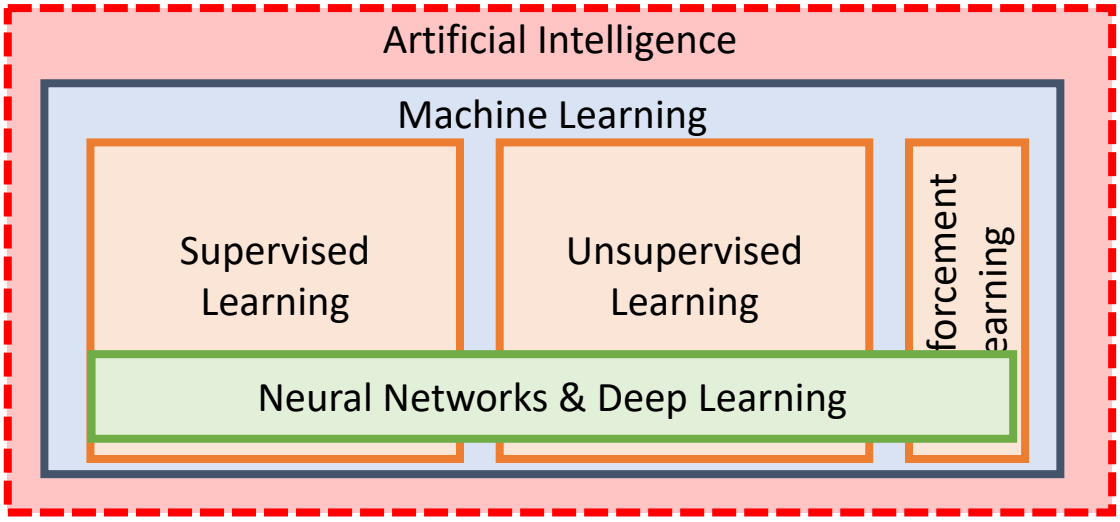

14

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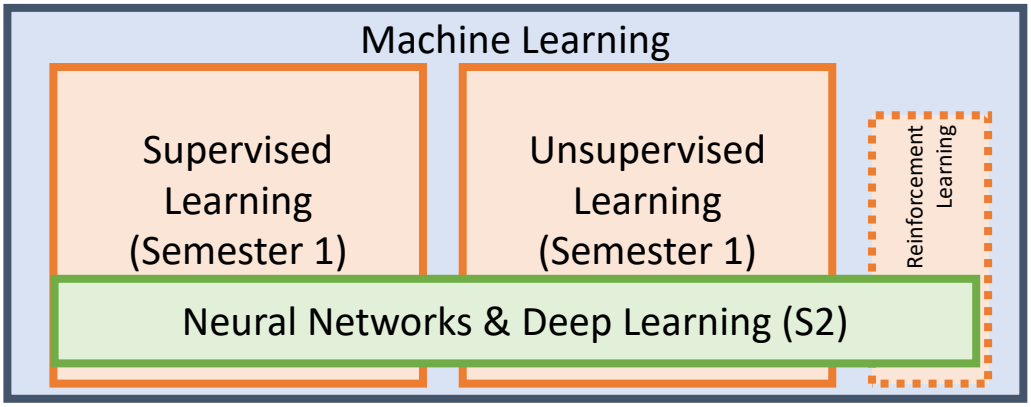

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Teaching Schedule (S1)

Week No.	Lecture A	Lecture B	Lab	Lecturer
1	Supervised Learning			Hongping
2				
3				
4				
5	External Considerations		Coursework	Georgios
6			Coursework	
7	Unsupervised Learning			
8				
9				
10				
11				
Christmas				

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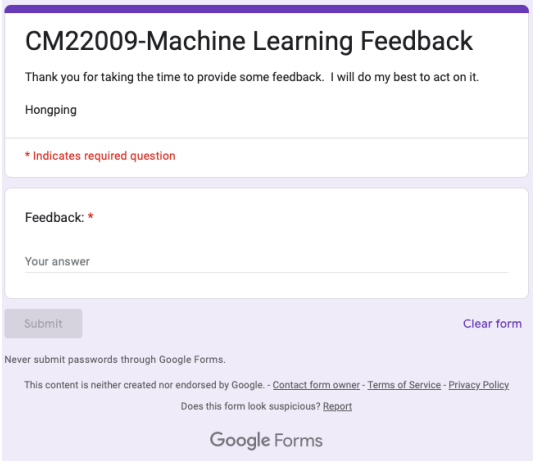
Feedback

- Feel free to leave feedback throughout the course.
- The QR code is the same each week.
- I monitor it weekly.
- Any feedback is good feedback so please let me know your thoughts.

SCAN ME



https://forms.gle/AyTC145yPeFVMiQw8

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Supervised Learning

Dr. Hongping Cai

CM22009 – Machine Learning

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Where are we?

Week No.	Lecture A	Lecture B
1 (29/09/2025)	Intro. Supervised Learning	Logistic Regression
2 (06/10/2025)	Nearest-Neighbours	Support Vector Machines
3 (13/10/2025)	Decision Trees	Random Forests
4 (20/10/2025)	Naïve-Bayes	Data Engineering
5 (27/10/2025)		
6 (03/11/2025)	Reading Week	

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Learning Objectives

Define the term supervised learning.

Describe the main components of a supervised learning method.

Illustrate the operation of the linear regression algorithm.

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Types of Learning

Supervised Learning

Unsupervised Learning

Reinforcement Learning

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Types of Learning

Supervised Learning

Learning through examples (with labels)

Cat

Dog

Unsupervised Learning

Reinforcement Learning

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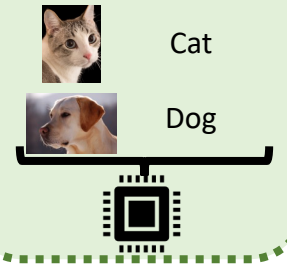
Types of Learning

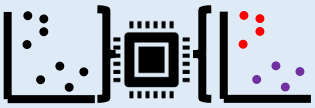
Supervised Learning

Learning through examples (with labels)



Unsupervised Learning

Learn the structure of the data (without labels)



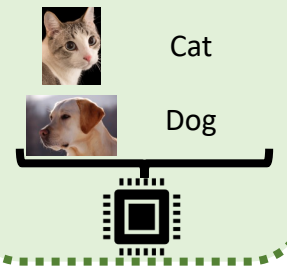
Reinforcement Learning

23

Types of Learning

Supervised Learning

Learning through examples (with labels)



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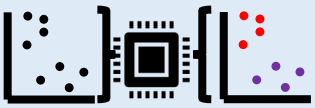
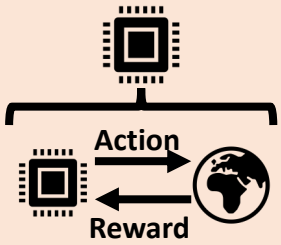


Reinforcement Learning

Learn by trial and error.



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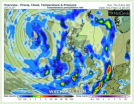
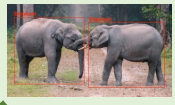
Types of Learning

Supervised Learning



Rainfall Forecasting

Classifying Fraudulent Transactions



Object Detection

Unsupervised Learning

Reinforcement Learning

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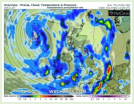
Types of Learning

Supervised Learning



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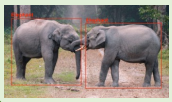
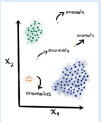


Object Detection

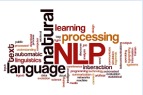
Unsupervised Learning

Customer Segmentation



Anomaly Detection

NLP



Reinforcement Learning

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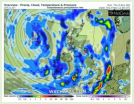
Types of Learning

Supervised Learning



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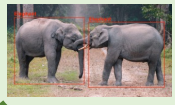

Object Detection

Unsupervised Learning

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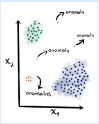


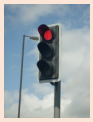
Anomaly Detection

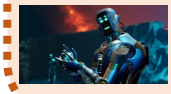
NLP

Reinforcement Learning

Personalised Learning

Traffic Control





Gaming AI

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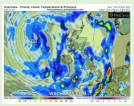
Types of Learning

Supervised Learning



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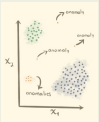

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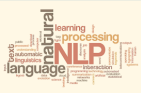

Reinforcement Learning

Personalised Learning

Traffic Control

Gaming AI

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Creating a Supervised Learning Model

- In order create a model we need to select a few things:

Type of Model

The type of model we choose will determine how we make predictions.

Cost Function

The cost-function determines how good our model is based on the predictions it makes.

Optimizer

The optimizer determines how we modify and improve our model based on the results of a cost function.

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Creating a Supervised Learning Model

- In order create a model we need to select a few things:

Type of Model **Cost Function** **Optimizer**

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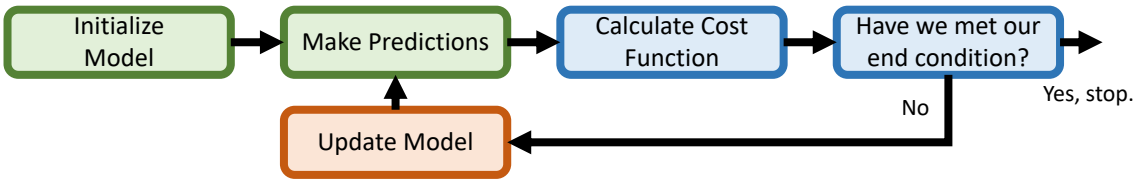
Creating a Supervised Learning Model

- In order create a model we need to select a few things:

Type of Model | **Cost Function** | **Optimizer**

- In general, the training works as follows:



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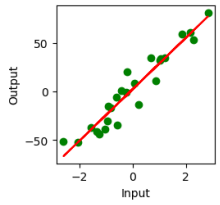
Linear Regression

- Linear regression is a regression model which means it aims to predict continuous values.
  - Next lecture we'll look at a classification variant.
- Essentially what linear regression does is try to find the line/plane/hyperplane of best fit through some data.



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Linear Regression

- To do this, Linear regression tries to optimise a hypothesis function.
- This hypothesis function follows the same format as the equation of a line:

$$h_\theta(x) = \theta_0 + \theta_1 x$$

33

Linear Regression

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$$\boxed{h_\theta(x)} = \boxed{\theta_0} + \boxed{\theta_1}\boxed{x}$$

Hypothesis Function

Parameter 0

Parameter 1

Input Value

34

Linear Regression

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- This hypothesis function follows the same format as the equation of a line:

$$\boxed{h_\theta(x)} = \boxed{\theta_0} + \boxed{\theta_1}\boxed{x}$$

Hypothesis Function

Input Value

Parameter 0

Parameter 1

- Look familiar? (y=mx+c)
- This is known as a **hypothesis function** because it is a proposed mapping, not necessarily the best mapping.

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Linear Regression

- Let's visualise this through an example:

"A gem merchant wants to create a model to predict the price of a gem based on its weight and purity."

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

Inputs/Features

Output/Truth

36

Linear Regression

- Let's visualise this through an example:

"A gem merchant wants to create a model to predict the price of a gem based on its weight and purity."

- The hypothesis function for this model will be as follows:

$$h_\theta(x) = \boxed{\theta_0} + \boxed{\theta_1 x_{weight} + \theta_2 x_{purity}}$$

And we need our bias parameter here, to shift the trend to the necessary location.

Two features therefore we need two parameters.

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

Inputs/Features

Output/Truth

37

Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

- Let's vectorize this:

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

38

Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

- Let's vectorize this:

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad \boldsymbol{\theta} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}$$

Weight	Purity	Price
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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

- Let's vectorize this:

$$\underbrace{\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad \boldsymbol{\theta} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}}_{\text{Dimension mis-match}}$$

Weight	Purity	Price
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40

Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

- Let's vectorize this:

$$\cancel{\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} \quad \boldsymbol{x} = \begin{bmatrix} 1 \\ x_1 \\ x_2 \end{bmatrix} \boldsymbol{\theta} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}$$

Dimension mis-match

Weight	Purity	Price
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Linear Regression

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2	20	40
2.5	80	120

$$h_\theta(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$$

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Linear Regression

- Current hypothesis function:

$$\cancel{h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2}$$

- Let's vectorize this:

$$\cancel{\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} \quad \boldsymbol{x} = \begin{bmatrix} 1 \\ x_1 \\ x_2 \end{bmatrix} \boldsymbol{\theta} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}$$

Dimension mis-match

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

$$h_\theta(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$$

$$h_\theta(x) = \boldsymbol{\theta}^{\boldsymbol{T}} \boldsymbol{x}$$

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^{\boldsymbol{T}} \boldsymbol{x}$$

- Example:
  - Let's use $\boldsymbol{x^0} = \{5, 100\}$
  - Let's set some easy parameters: $\boldsymbol{\theta} = \{2,2,2\}$

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$$

- Example:
  - Let's use $\boldsymbol{x^0} = \{5, 100\}$
  - Let's set some easy parameters: $\boldsymbol{\theta} = \{2,2,2\}$
  - Modify $\boldsymbol{x^0}$ to match parameters: $\boldsymbol{x^0} = \{1, 5, 100\}$

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$$

- Example:
  - Let's use $\boldsymbol{x^0} = \{5, 100\}$
  - Let's set some easy parameters: $\boldsymbol{\theta} = \{2, 2, 2\}$
  - Modify $\boldsymbol{x^0}$ to match parameters: $\boldsymbol{x^0} = \{1, 5, 100\}$
  - Calculate hypothesis:

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

$$h_\theta(\boldsymbol{x^0}) = \boldsymbol{\theta}^T \boldsymbol{x} = [2 \quad 2 \quad 2] \begin{bmatrix} 1 \\ 5 \\ 100 \end{bmatrix} = \sum \begin{bmatrix} 2 \\ 10 \\ 200 \end{bmatrix} = 212$$

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$$

- Example:
  - Let's use $\boldsymbol{x^0} = \{5, 100\}$
  - Let's set some easy parameters: $\boldsymbol{\theta} = \{2, 2, 2\}$
  - Modify $\boldsymbol{x^0}$ to match parameters: $\boldsymbol{x^0} = \{1, 5, 100\}$
  - Calculate hypothesis: 212

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$$

- Example:
  - Let's use $\boldsymbol{x^0} = \{5, 100\}$
  - Let's set some easy parameters: $\boldsymbol{\theta} = \{2, 2, 2\}$
  - Modify $\boldsymbol{x^0}$ to match parameters: $\boldsymbol{x^0} = \{1, 5, 100\}$
  - Calculate hypothesis: 212
  - Calculate cost for $\boldsymbol{x^0}$:

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

Mean-Squared Error

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^i) - y^i)^2$$

Half the mean of the squared error for all training examples.

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^{\boldsymbol{T}}\boldsymbol{x}$$

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

- Example:
  - Let's use $\boldsymbol{x^0} = \{5, 100\}$
  - Let's set some easy parameters: $\boldsymbol{\theta} = \{2, 2, 2\}$
  - Modify $\boldsymbol{x^0}$ to match parameters: $\boldsymbol{x^0} = \{1, 5, 100\}$
  - Calculate hypothesis: 212
  - Calculate cost for $\boldsymbol{x^0}$:

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^i) - y^i)^2$$

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^{\boldsymbol{T}}\boldsymbol{x}$$

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

- Example:
  - Let's use $\boldsymbol{x^0} = \{5, 100\}$
  - Let's set some easy parameters: $\boldsymbol{\theta} = \{2, 2, 2\}$
  - Modify $\boldsymbol{x^0}$ to match parameters: $\boldsymbol{x^0} = \{1, 5, 100\}$
  - Calculate hypothesis: 212
  - Calculate cost for $\boldsymbol{x^0}$:

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^i) - y^i)^2 = (h_\theta(x^0) - y^0)^2$$

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$$

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

- Example:
  - Let's use $\boldsymbol{x^0} = \{5, 100\}$
  - Let's set some easy parameters: $\boldsymbol{\theta} = \{2, 2, 2\}$
  - Modify $\boldsymbol{x^0}$ to match parameters: $\boldsymbol{x^0} = \{1, 5, 100\}$
  - Calculate hypothesis for $\boldsymbol{x^0}$: 212
  - Calculate cost for $\boldsymbol{x^0}$:

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^i) - y^i)^2 = (h_\theta(x^0) - y^0)^2 = (212 - 500)^2 = 41472$$

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$$

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

- Example:
  - Let's use $\boldsymbol{x^0} = \{5, 100\}$
  - Let's set some easy parameters: $\boldsymbol{\theta} = \{2, 2, 2\}$
  - Modify $\boldsymbol{x^0}$ to match parameters: $\boldsymbol{x^0} = \{1, 5, 100\}$
  - Calculate hypothesis for $\boldsymbol{x^0}$: 212
  - Calculate cost for $\boldsymbol{x^0}$: 41472

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$$

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

- Example:
  - Now let's do the whole table.
  - Start by calculating the predictions for each (hypothesis values):

$$h_\theta(\boldsymbol{x^0}) = \mathbf{212}$$

$$h_\theta(\boldsymbol{x^1}) = \begin{bmatrix} 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 20 \end{bmatrix}$$

$$h_\theta(\boldsymbol{x^2}) = \begin{bmatrix} 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2.5 \\ 80 \end{bmatrix}$$

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$$

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

- Example:
  - Now let's do the whole table.
  - Start by calculating the predictions for each (hypothesis values):

$$h_\theta(x^0) = 212$$

$$h_\theta(x^1) = 46$$

$$h_\theta(x^2) = 167$$

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$$

- Example:
  - Now let's do the whole table.
  - Start by calculating the predictions for each (hypothesis values):

Weight	Purity	Price	Predictions
5	100	500	212
2	20	40	46
2.5	80	120	167

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$$

- Example:
  - Now let's do the whole table.
  - Start by calculating the predictions for each (hypothesis values).
  - Next we can calculate the error (cost) using $J(\theta)$:

Weight	Purity	Price	Predictions
5	100	500	212
2	20	40	46
2.5	80	120	167

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$$

Weight	Purity	Price	Predictions
5	100	500	212
2	20	40	46
2.5	80	120	167

- Example:
  - Now let's do the whole table.
  - Start by calculating the predictions for each (hypothesis values).
  - Next we can calculate the error (cost) using $J(\theta)$:

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^i) - y^i)^2$$

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$$

Weight	Purity	Price	Predictions
5	100	500	212
2	20	40	46
2.5	80	120	167

- Example:
  - Now let's do the whole table.
  - Start by calculating the predictions for each (hypothesis values).
  - Next we can calculate the error (cost) using $J(\theta)$:

$$J(\theta) = \frac{1}{2(3)}((212 - 500)^2 + (46 - 40)^2 + (167 - 120)^2)$$

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$$

Weight	Purity	Price	Predictions
5	100	500	212
2	20	40	46
2.5	80	120	167

- Example:
  - Now let's do the whole table.
  - Start by calculating the predictions for each (hypothesis values).
  - Next we can calculate the error (cost) using $J(\theta)$:

$$J(\theta) = \frac{1}{2(3)}(82944 + 36 + 2209)$$

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$$

Weight	Purity	Price	Predictions
5	100	500	212
2	20	40	46
2.5	80	120	167

- Example:
  - Now let's do the whole table.
  - Start by calculating the predictions for each (hypothesis values).
  - Next we can calculate the error (cost) using $J(\theta)$:

$$J(\theta) = \frac{1}{6}(85189)$$

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$$

Weight	Purity	Price	Predictions
5	100	500	212
2	20	40	46
2.5	80	120	167

- Example:
  - Now let's do the whole table.
  - Start by calculating the predictions for each (hypothesis values).
  - Next we can calculate the error (cost) using $J(\theta)$:

$$J(\theta) = 14198.167$$

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Linear Regression

- Current hypothesis function:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$$

Weight	Purity	Price	Predictions
5	100	500	212
2	20	40	46
2.5	80	120	167

- Example:
  - Now let's do the whole table.
  - Start by calculating the predictions for each (hypothesis values).
  - Next we can calculate the error (cost) using $J(\theta)$:

$$J(\theta) = 14198.167$$

  - We will then apply an optimisation algorithm to improve our parameters (θ) and iterate!
    - You will be doing this in the lab! ☺

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Gradient Descent

- Iterative optimization algorithm
- Uses repeating steps in the 'opposite' direction of the gradient.
- We'll use this to find optimal parameters ($\boldsymbol{\theta}$)

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Gradient Descent

- Hypothesis function: $h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$
- Cost function: $J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^i) - y^i)^2$

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

- Iterative routine
- At each step perform this for each parameter:

$$\theta_j = \theta_j - \boxed{\alpha}\frac{\partial}{\partial \theta_j} J(\theta)$$

Learning Rate
(How fast do we want the parameters to change.)

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Gradient Descent

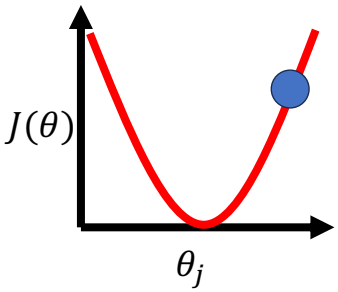
- Hypothesis function: $h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$
- Cost function: $J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^i) - y^i)^2$

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

- Iterative routine
- At each step perform this for each parameter:

$$\theta_j = \theta_j - \alpha \boxed{\frac{\partial}{\partial \theta_j} J(\theta)}$$

Inverse of the derivative of
the cost function w.r.t the parameter



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Gradient Descent

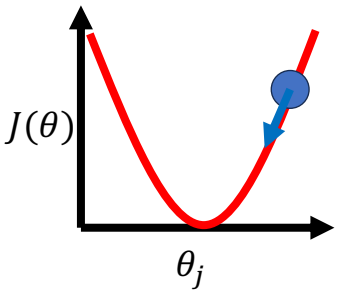
- Hypothesis function: $h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}$
- Cost function: $J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^i) - y^i)^2$

Weight	Purity	Price
5	100	500
2	20	40
2.5	80	120

- Iterative routine
- At each step perform this for each parameter:

$$\theta_j = \theta_j \boxed{-} \alpha \boxed{\frac{\partial}{\partial \theta_j} J(\theta)}$$

Inverse of the derivative of
the cost function w.r.t the parameter



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Summary: Linear Regression

- Hypothesis function: $h_\theta(x) = \theta^T x$
- Cost function: $J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^i) - y^i)^2$
- Gradient Descent: repeat this updating for each parameter:

$$\theta_j = \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta) = \theta_j - \alpha \frac{1}{m}\sum_{i=1}^{m}(h_\theta(x^i) - y^i)x_j^i$$

Stop when it reaches a certain cost threshold or number of iterations.

The derivation is on Moodle (Week 1)

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Ex: Gradient Descent

Use linear regression to model the price of three samples. Calculate the cost function of the first two iterations. Do you see the cost decrease?

- Hypothesis function: $h_\theta(x) = \boldsymbol{\theta^T x}$
- Cost function: $J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^i) - y^i)^2$
- Starting parameters: $\theta = \{2,2,2\}$
- Learning rate: 0.0001

Length	Width	Price	Iter 0	Iter 1
5	10	35		
2	50	156		
10	5	40		
		$J(\theta)$	?	?

Correct answer on next slide

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Ex: Gradient Descent

Length	Width	Price	Iter 0	Iter 1
5	10	35	32	32.890
2	50	156	106	108.212
10	5	40	32	32.482
		$J(\theta)$	428.833	390.777

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Feedback!

If you have any feedback, use the QR code or link to anonymously report it.

I'm open to any comments/feedback.

SCAN ME

https://forms.gle/AyTC145yPeFVMiQw8

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